

Lecture 22

Edmonds-Karp Algorithm (contd.)

Edmonds-Karp Algorithm

Edmonds-Karp(G, s, t):

1. **for** each edge $(u, v) \in E(G)$
2. $f(u, v) = 0$
3. **while** there exists an $s \rightsquigarrow t$ path in the residual network G_f
4. $P =$ shortest length $s \rightsquigarrow t$ path
5. $\delta = \text{Min}(c_f(u, v) : (u, v) \text{ in } P)$
6. **for** each edge (u, v) in P
7. **if** $(u, v) \in E$
8. $f(u, v) = f(u, v) + \delta$
9. **else**
10. $f(v, u) = f(v, u) - \delta$
11. **return** f

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Proof: We will do a proof by contradiction.

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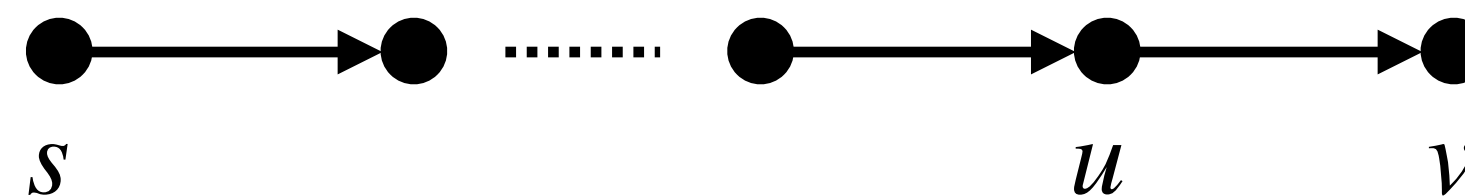
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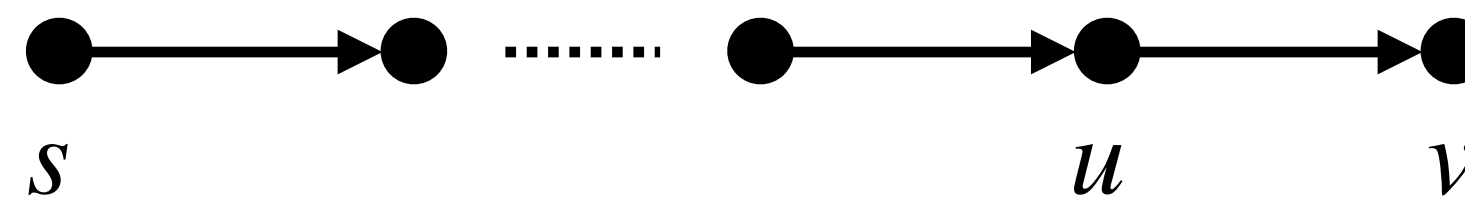
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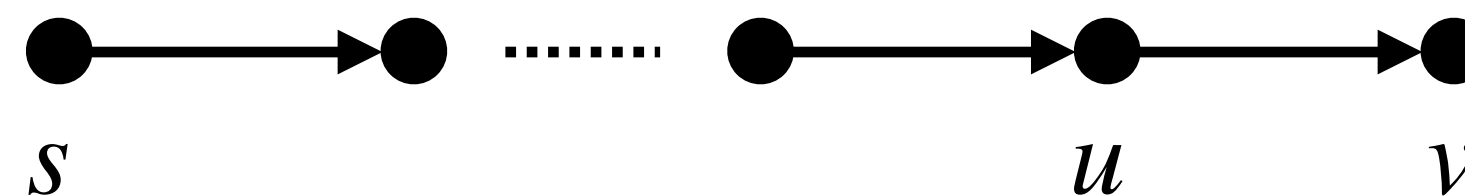
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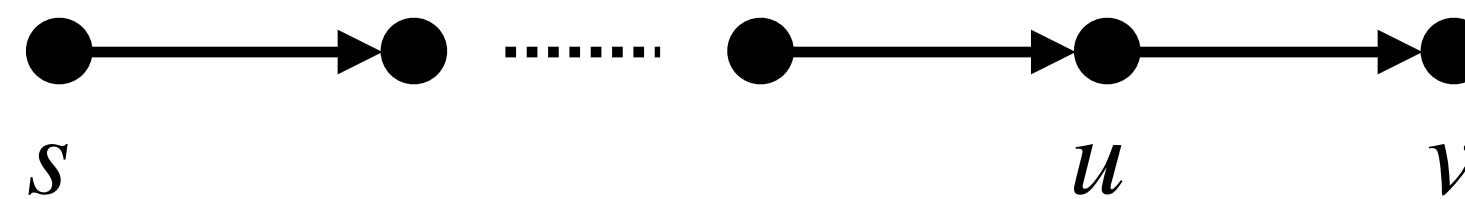
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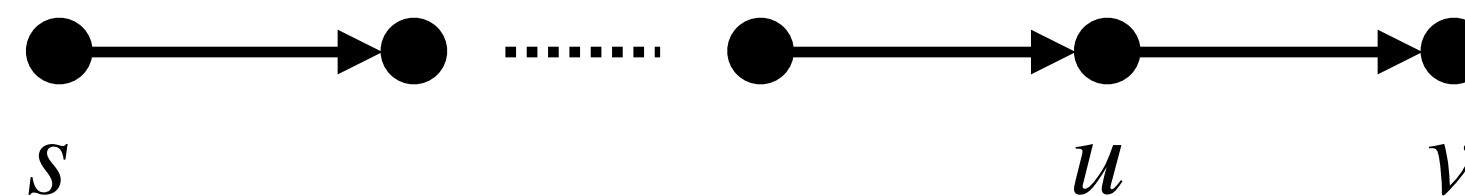
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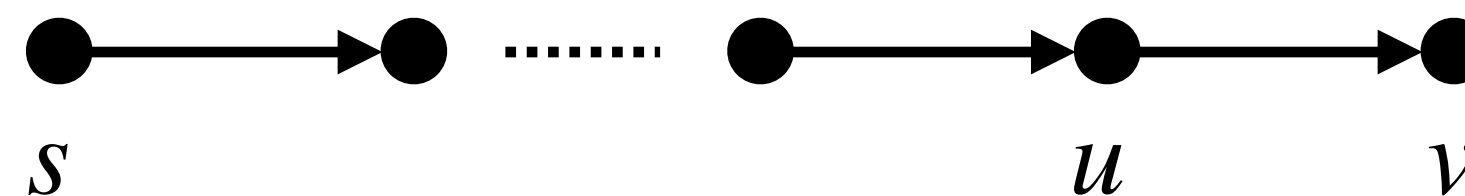
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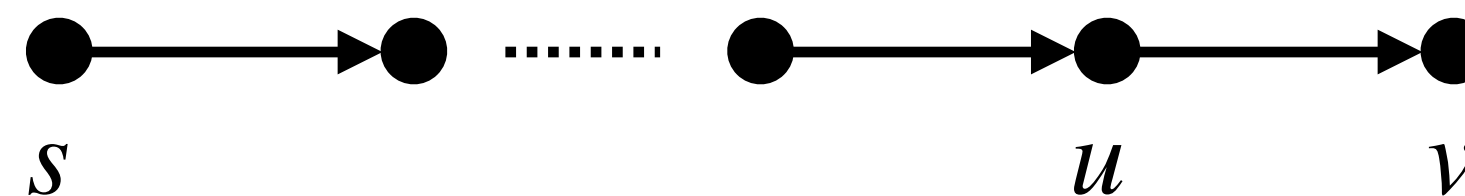
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Edmonds-Karp: Runtime Analysis

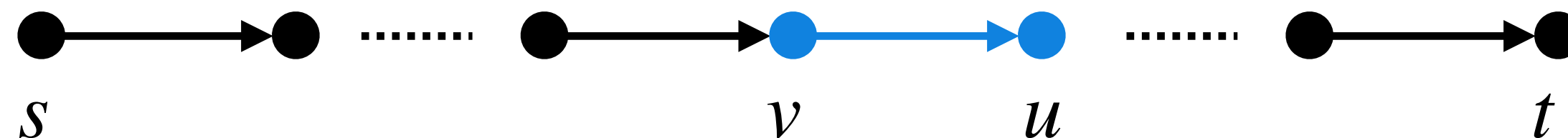
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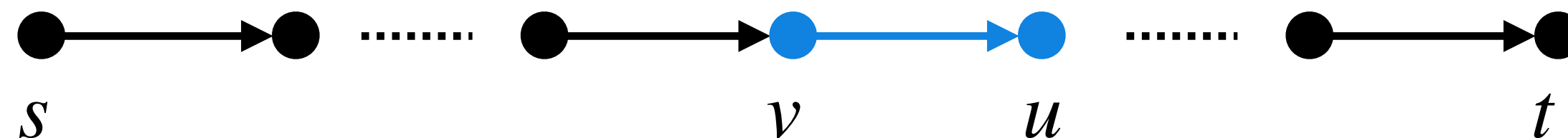
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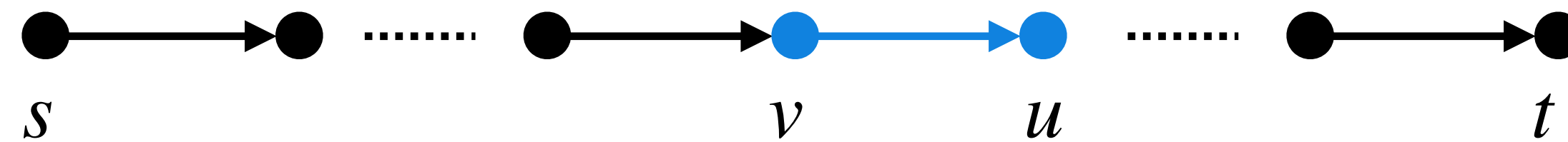
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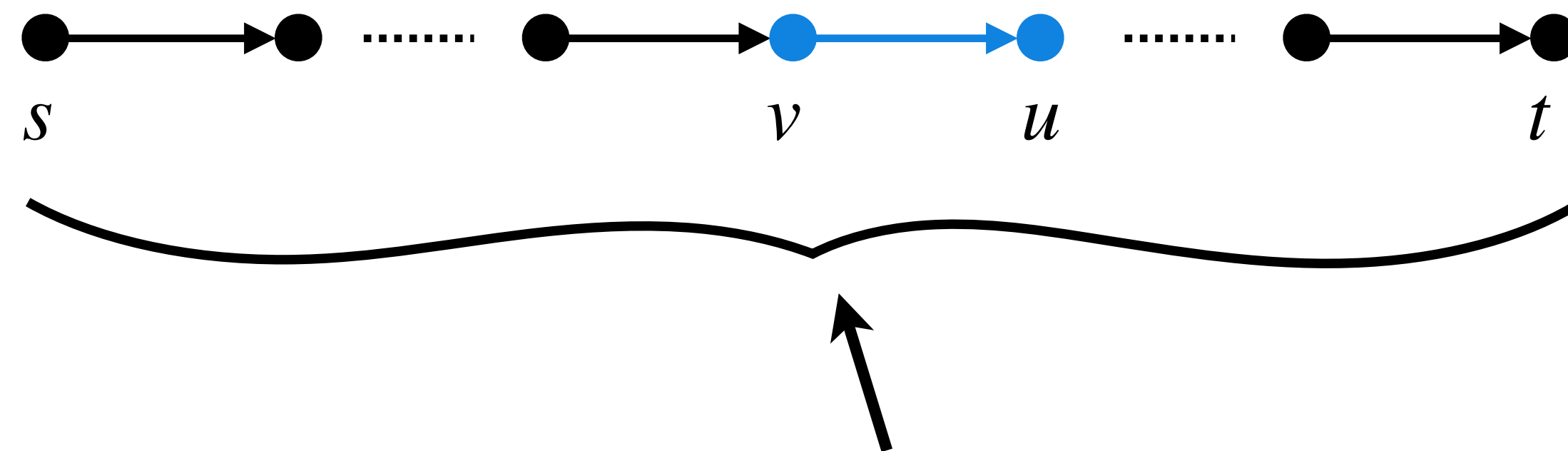
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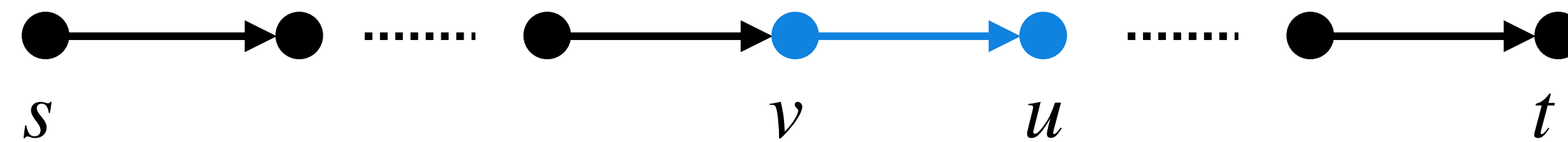


Augmenting path is a shortest path from s to t in G_f

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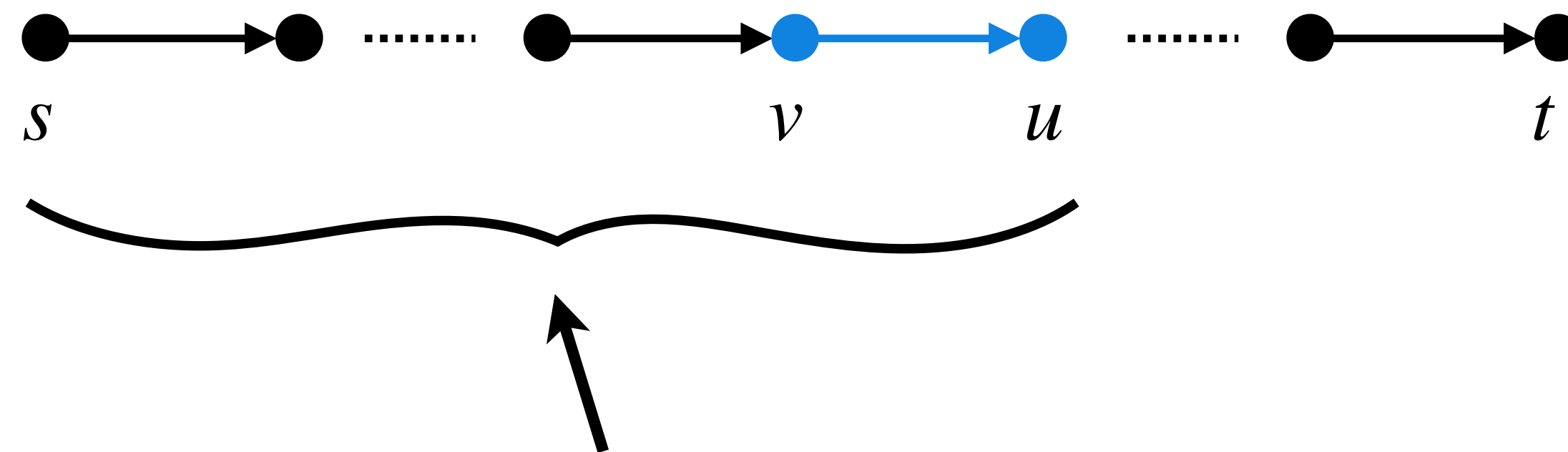
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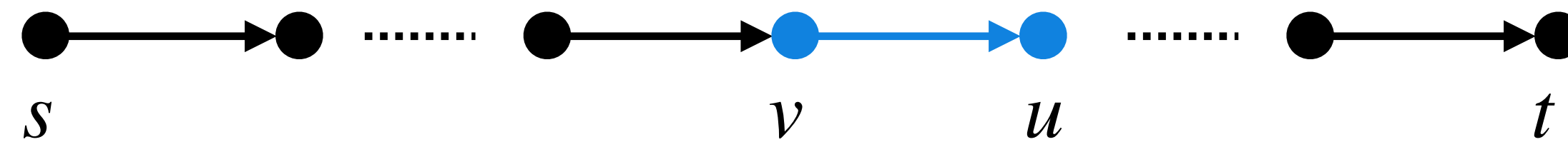


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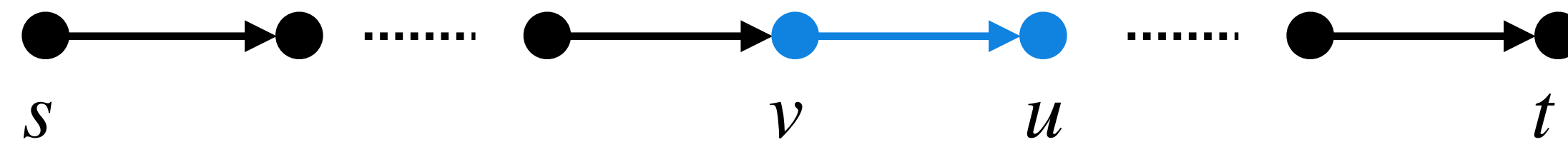
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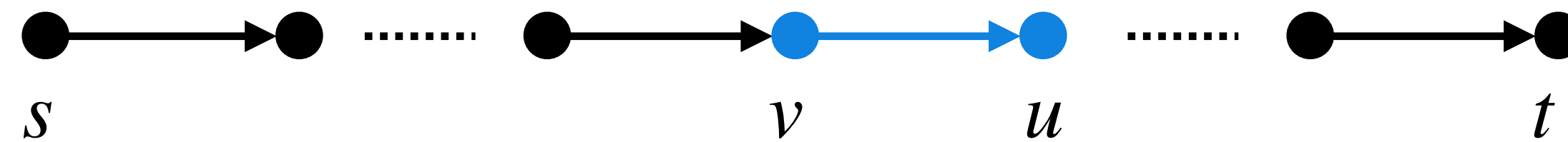


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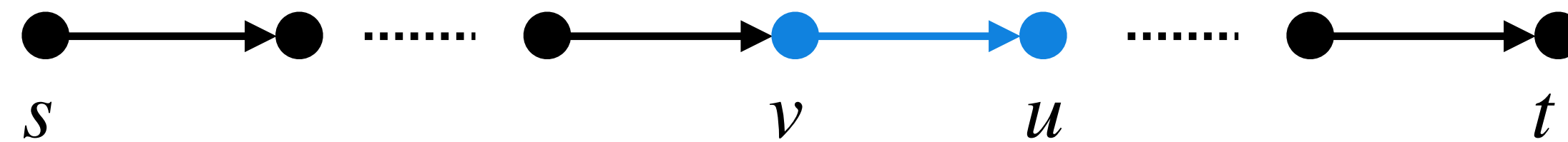
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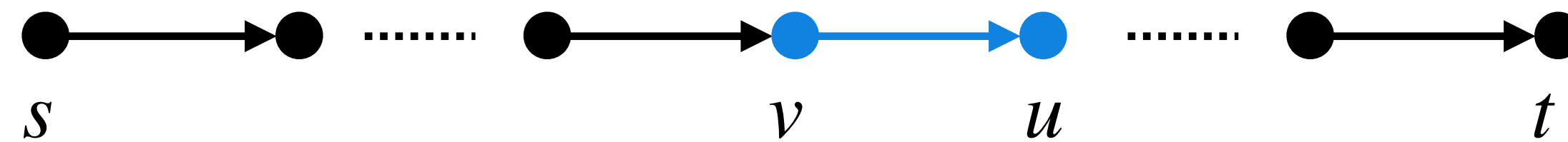
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$$\begin{aligned}\delta_f(s, v) &= \delta_f(s, u) - 1 \\ &\leq \delta_{f_1}(s, u) - 1\end{aligned}$$

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Augmentation must have decreased the flow from v to u . Hence, (v, u) must be a part of the **augmenting path** in G_f .



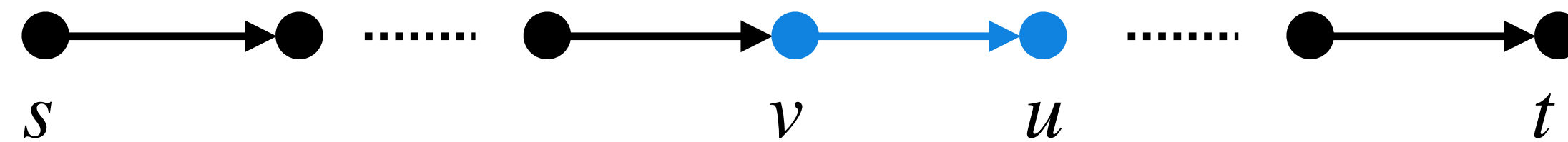
Therefore,

$$\begin{aligned}\delta_f(s, v) &= \delta_f(s, u) - 1 \\ &\leq \delta_{f_1}(s, u) - 1 \quad (\because \delta_f(s, u) \leq \delta_{f_1}(s, u))\end{aligned}$$

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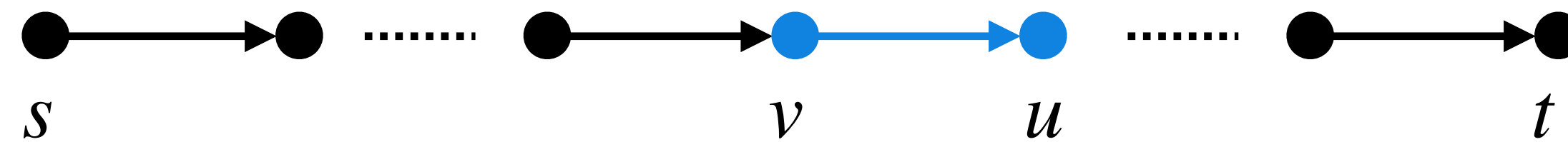
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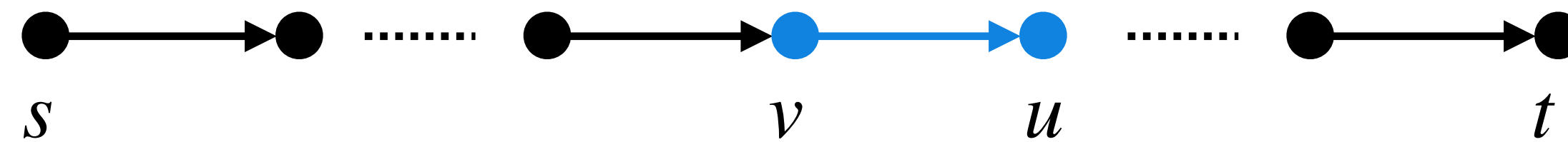
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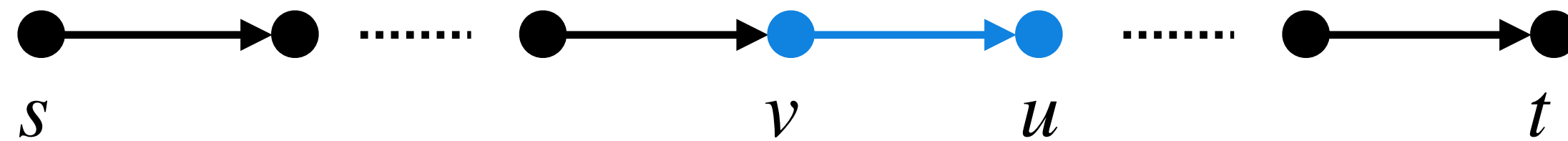
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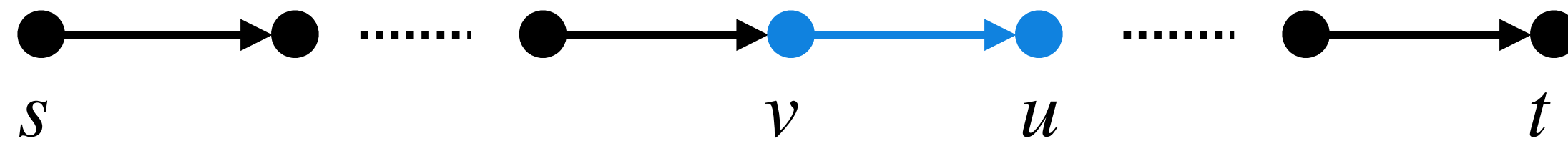
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Proof: On the next slide.

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Analysis when (u, v) is a backward edge is similar.

Edmonds-Karp: Runtime Analysis

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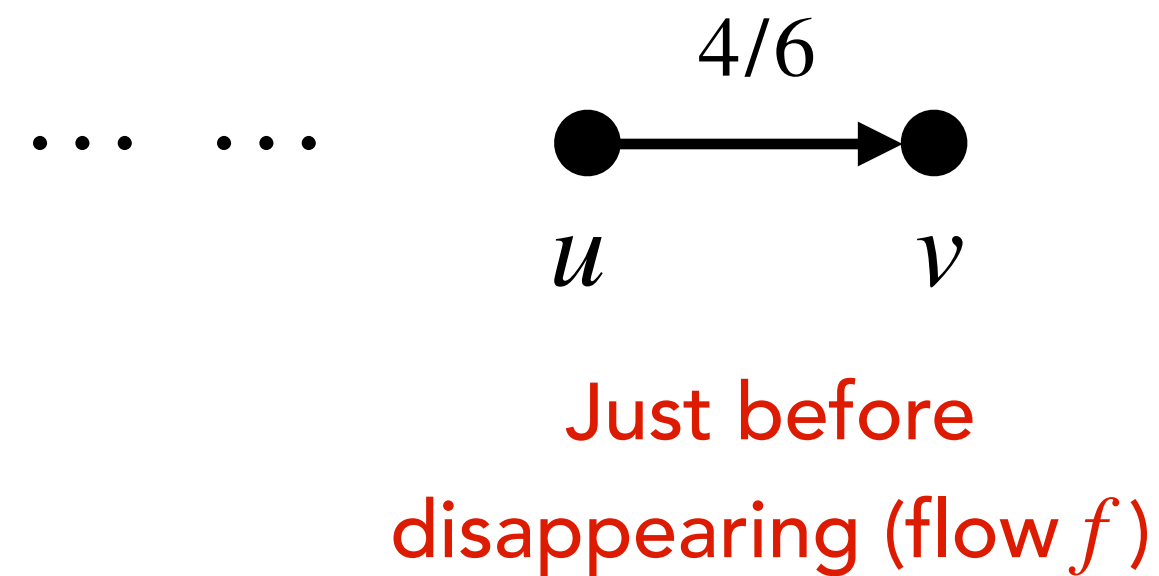
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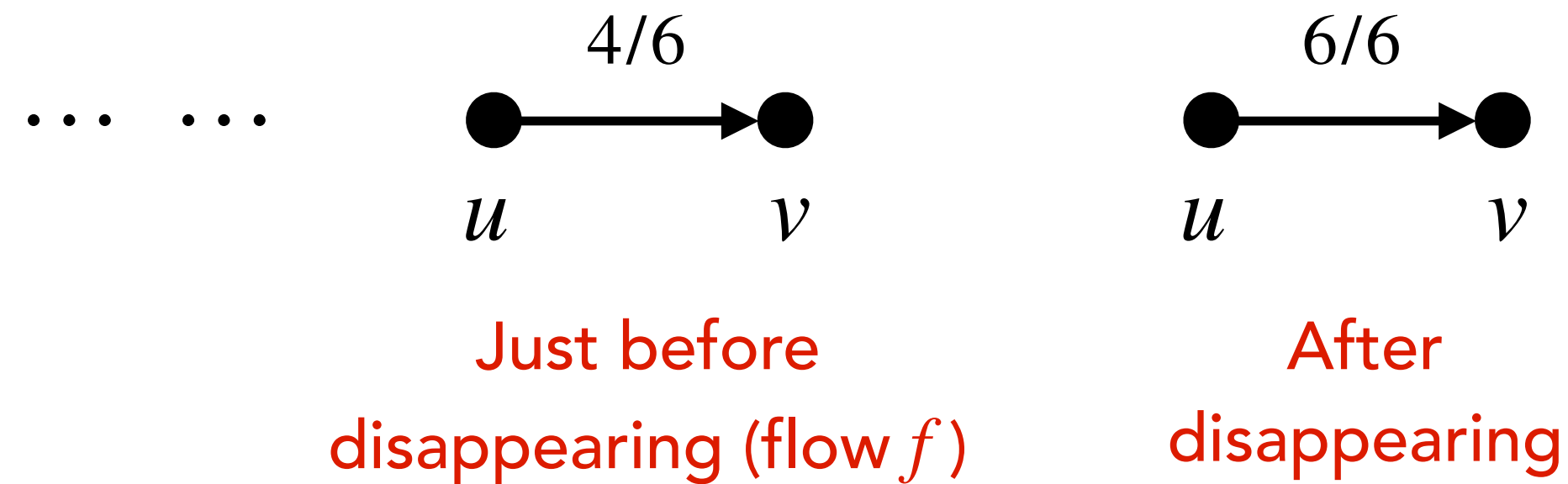
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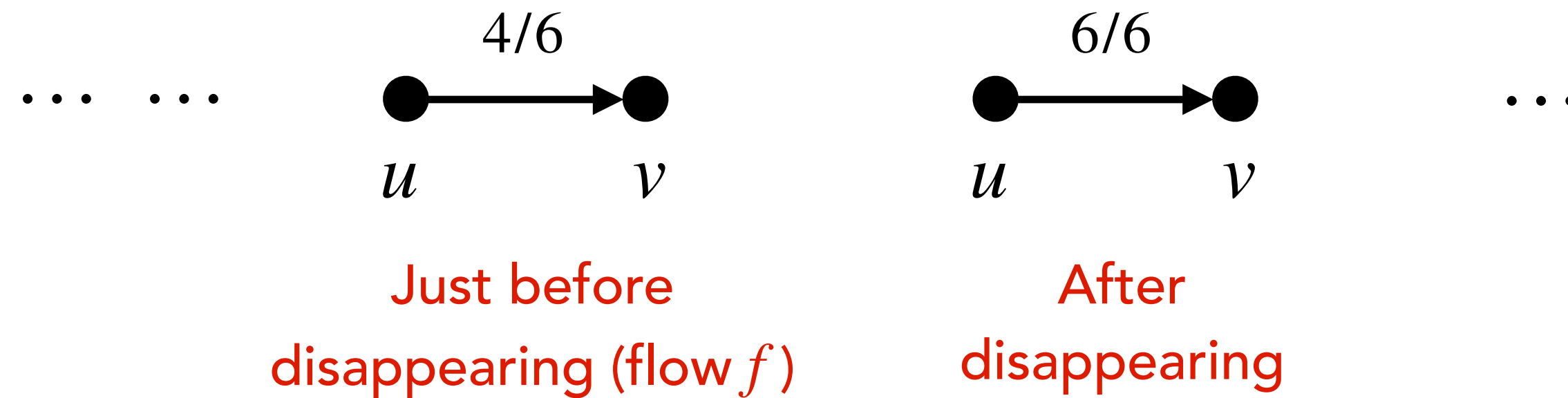
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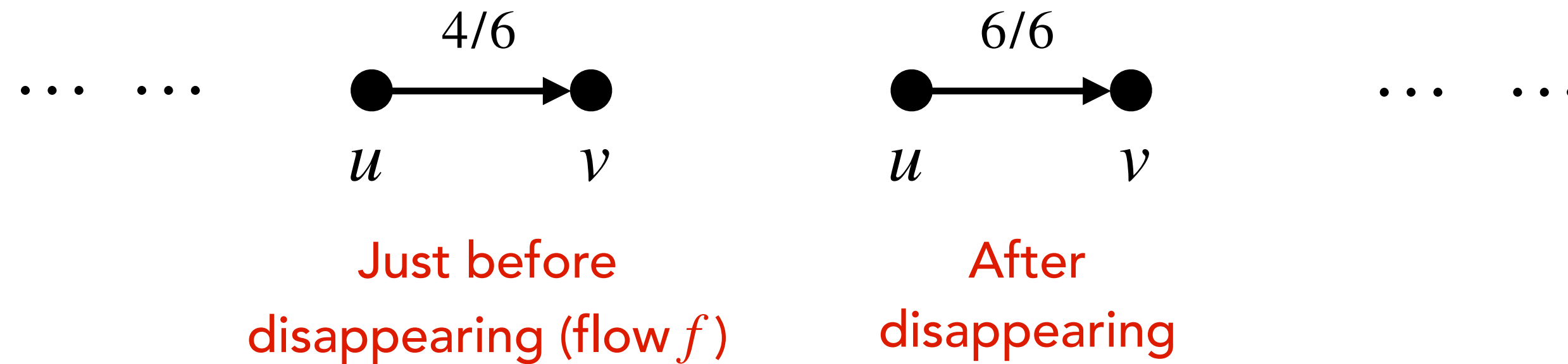
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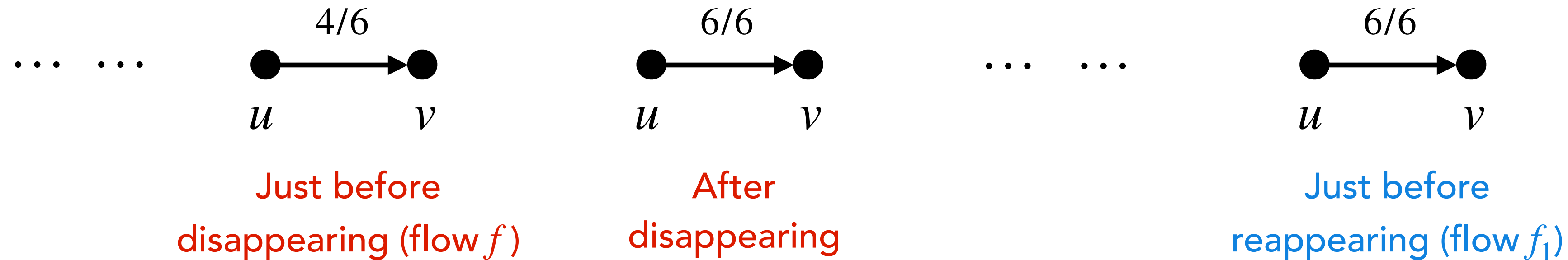
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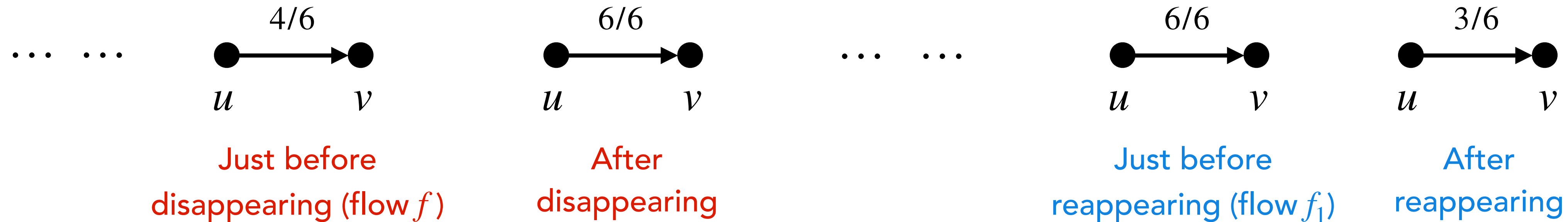
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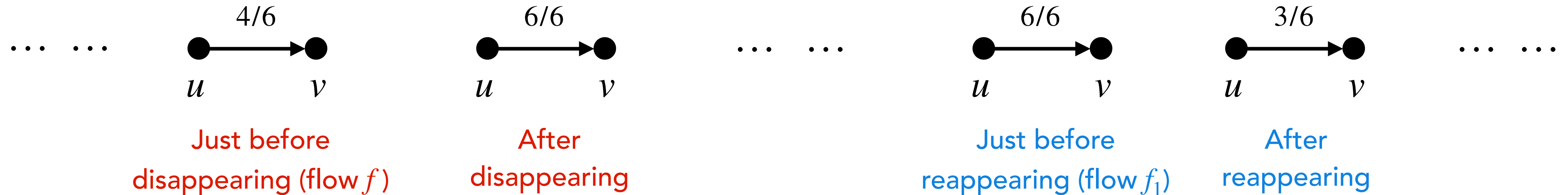
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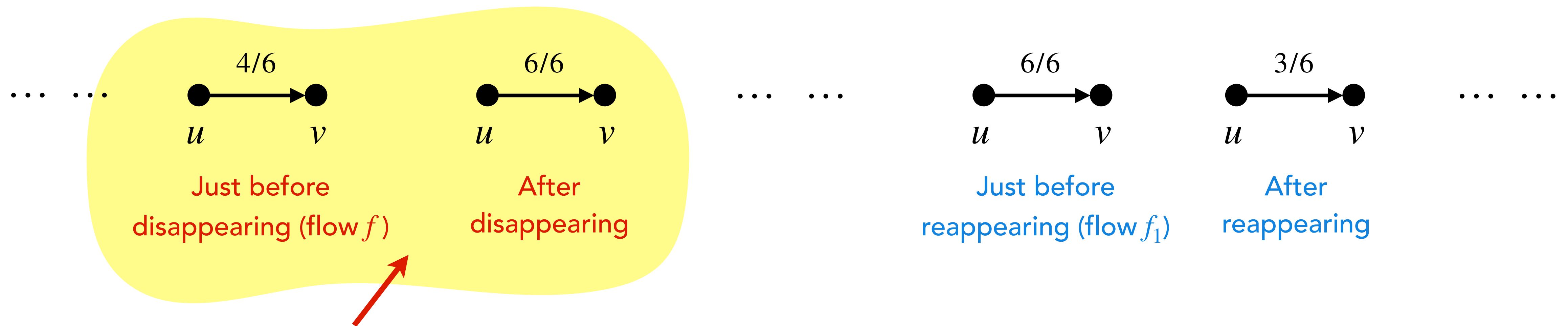
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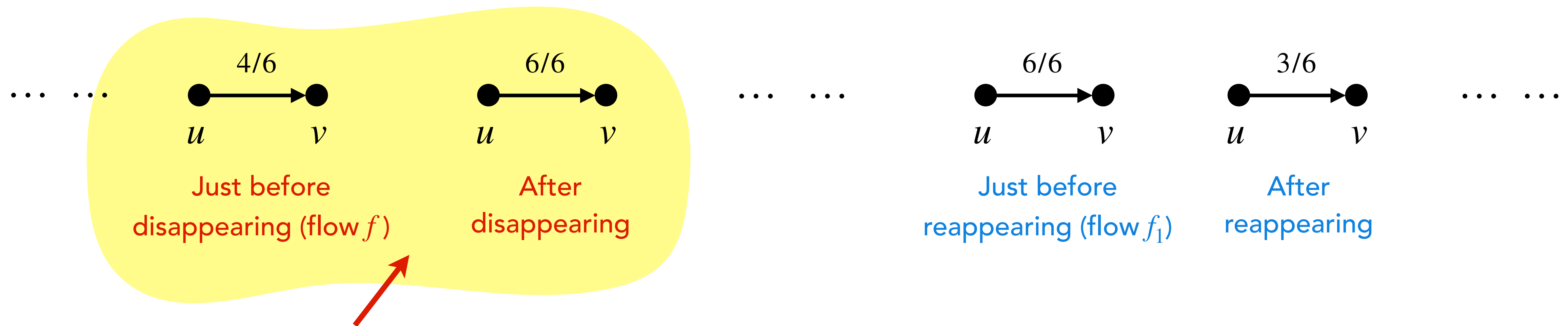


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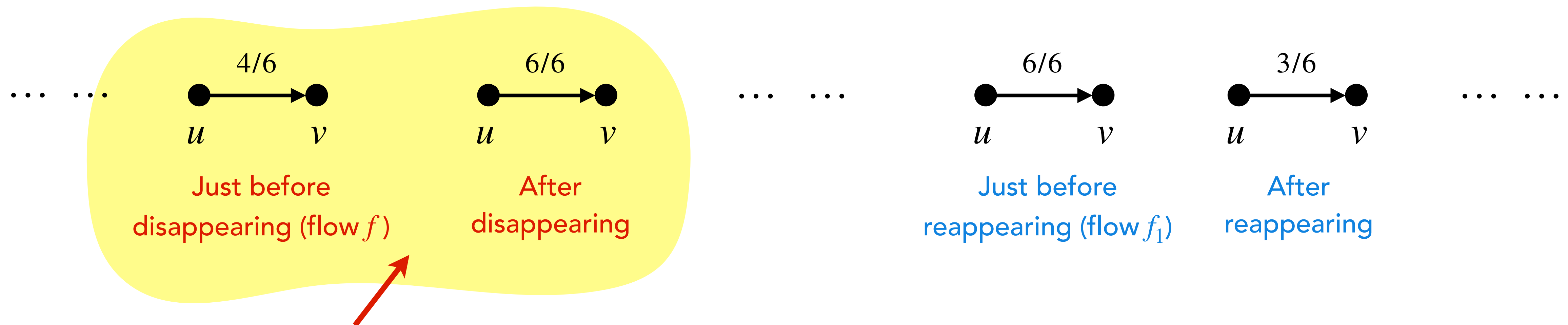
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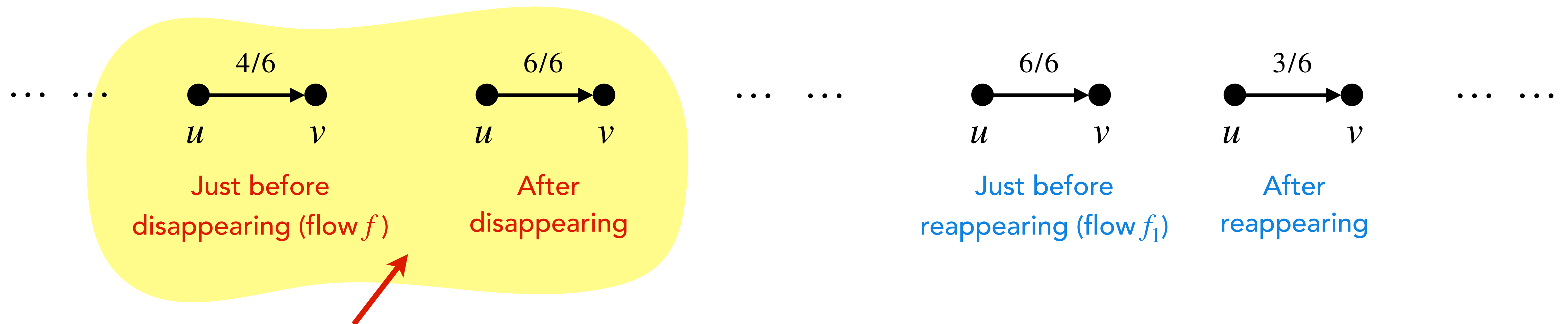
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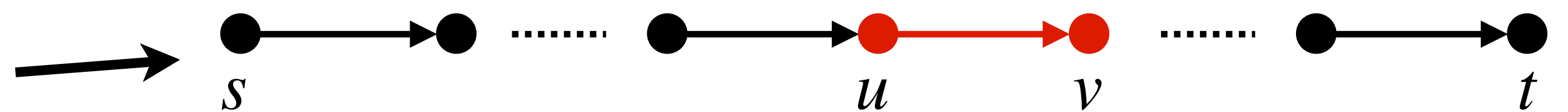
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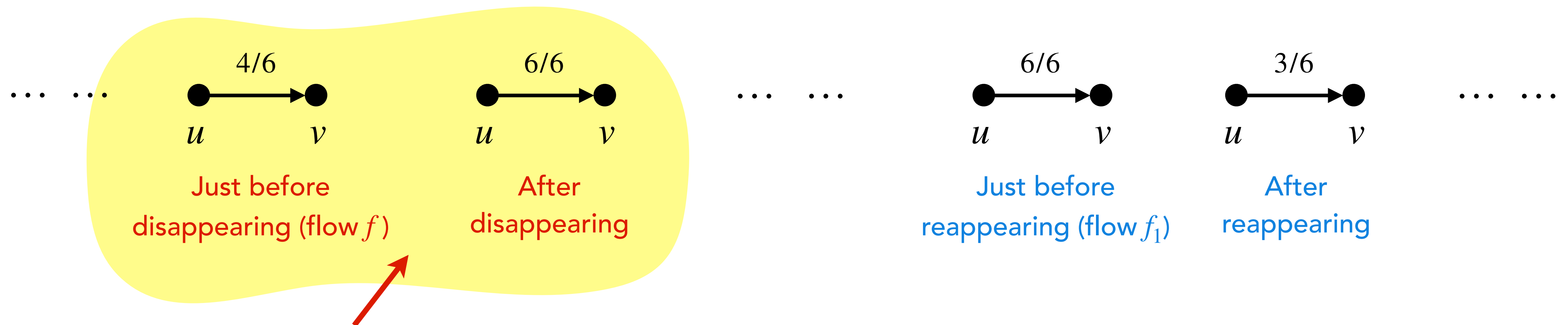
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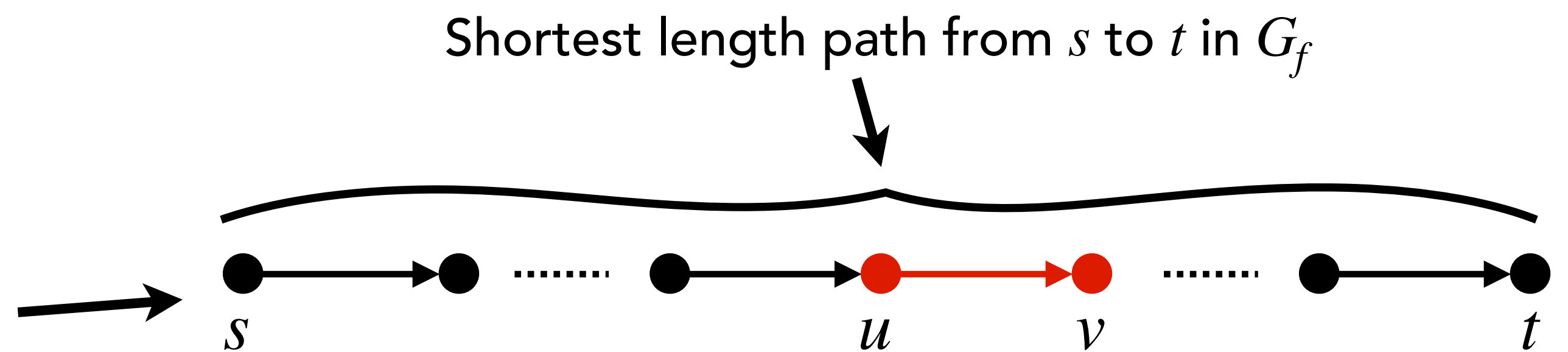
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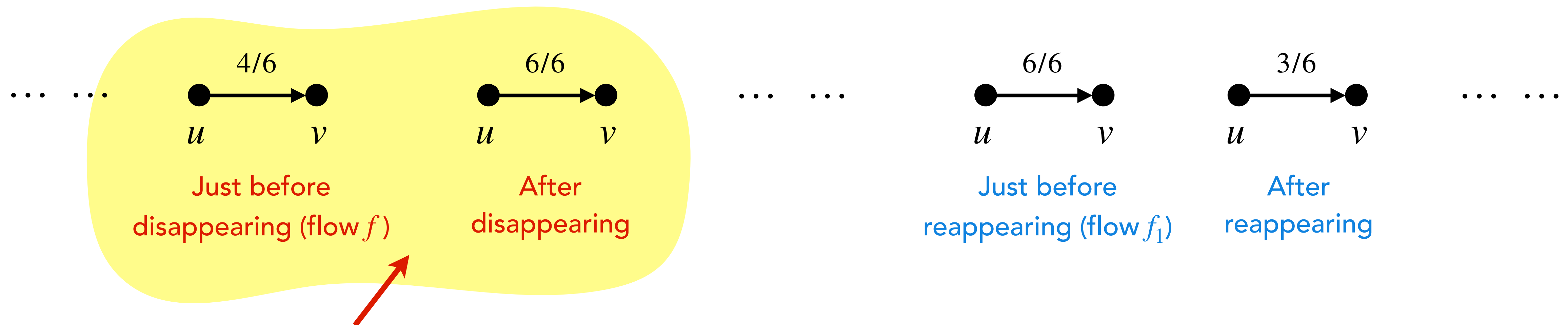
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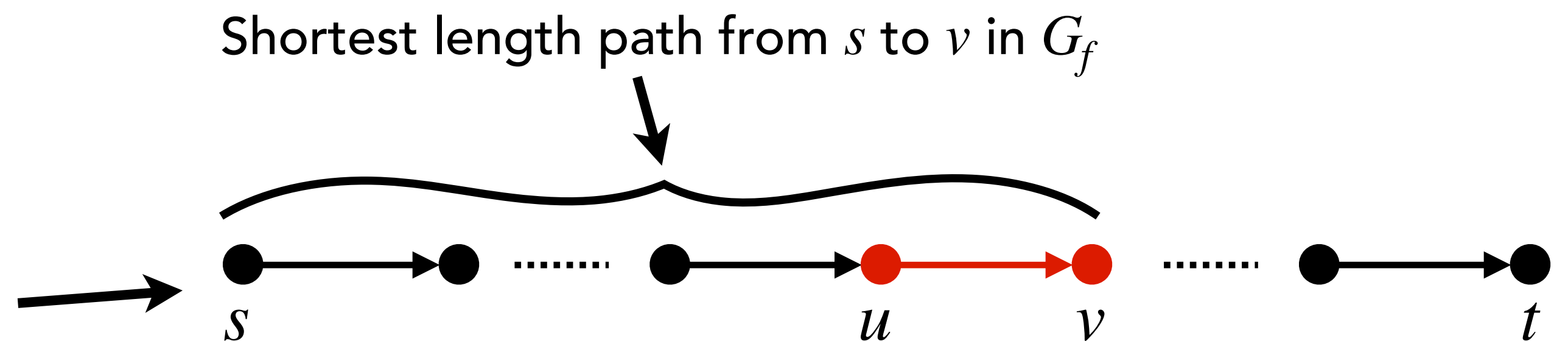
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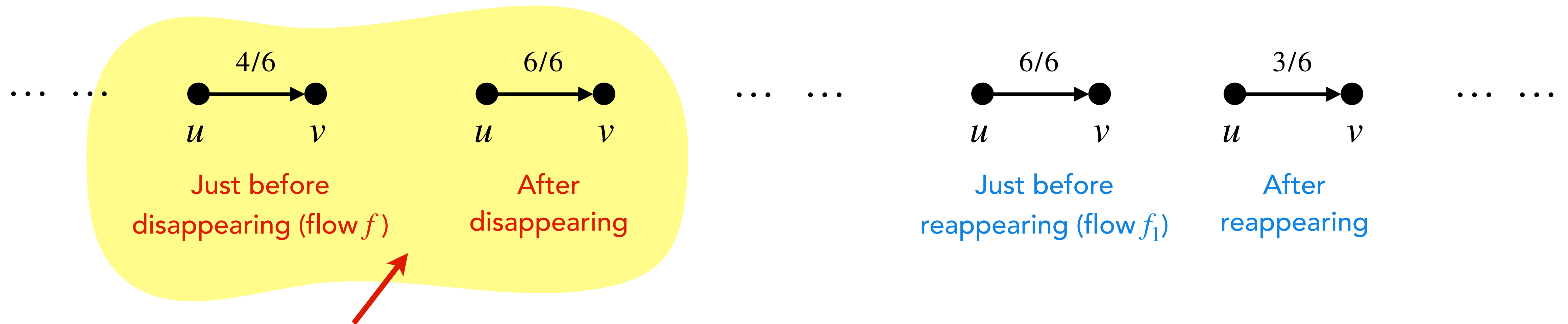
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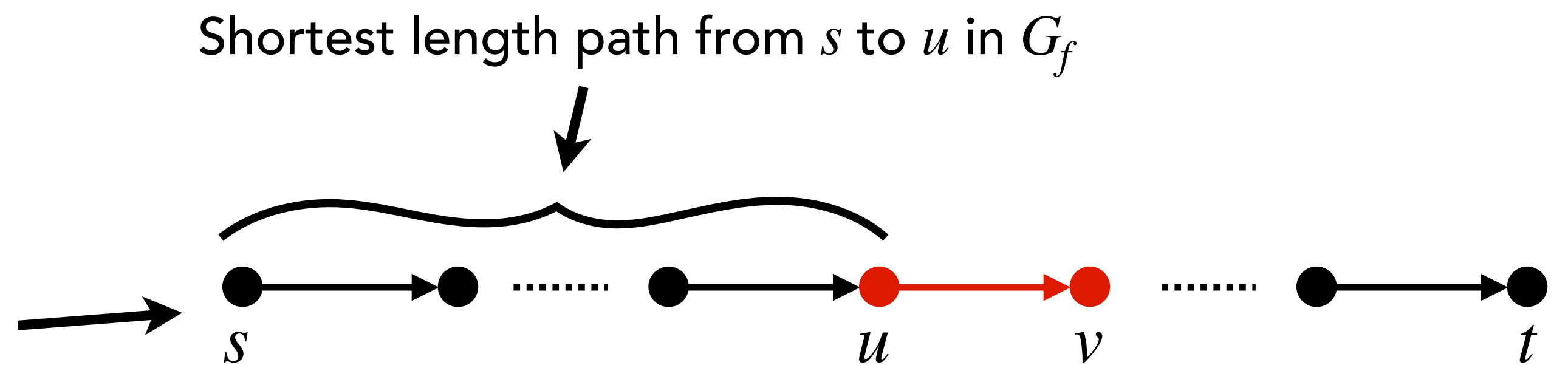
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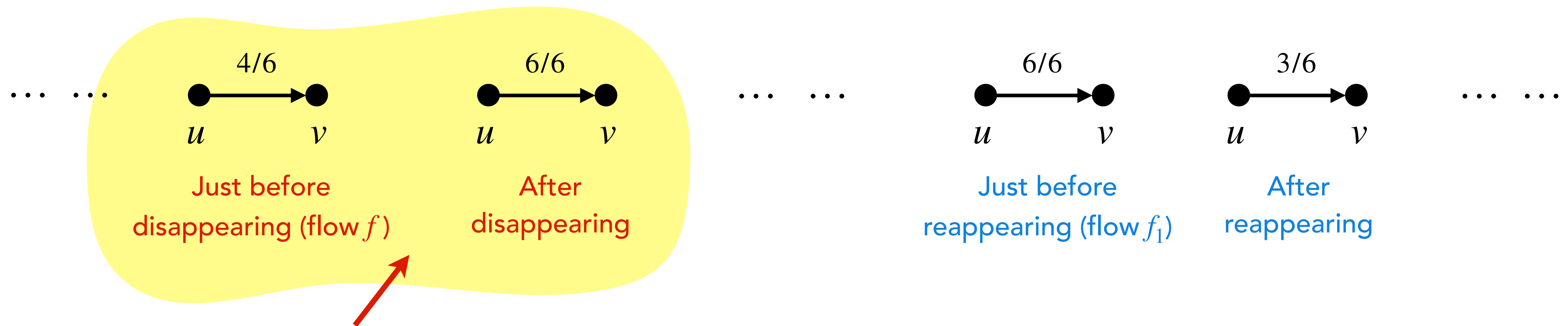
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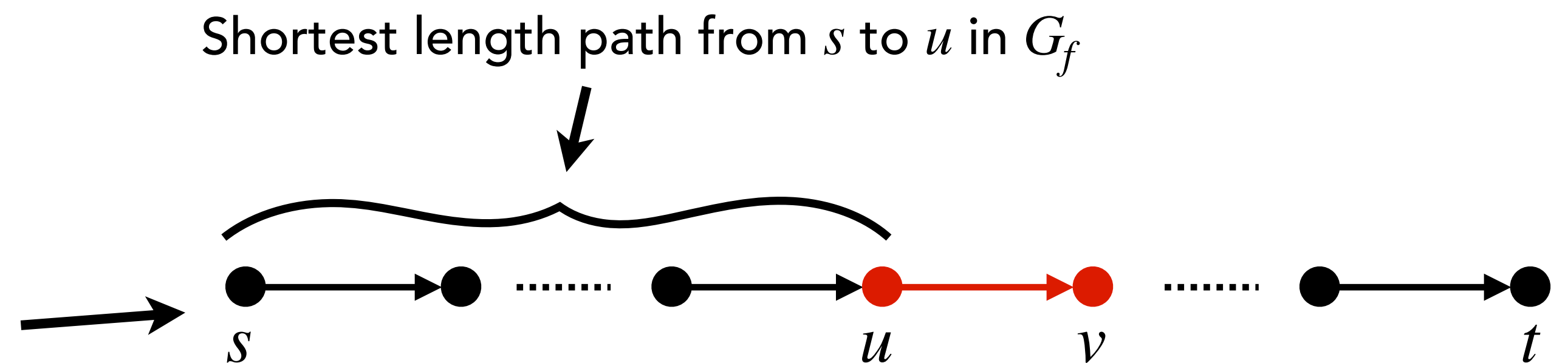
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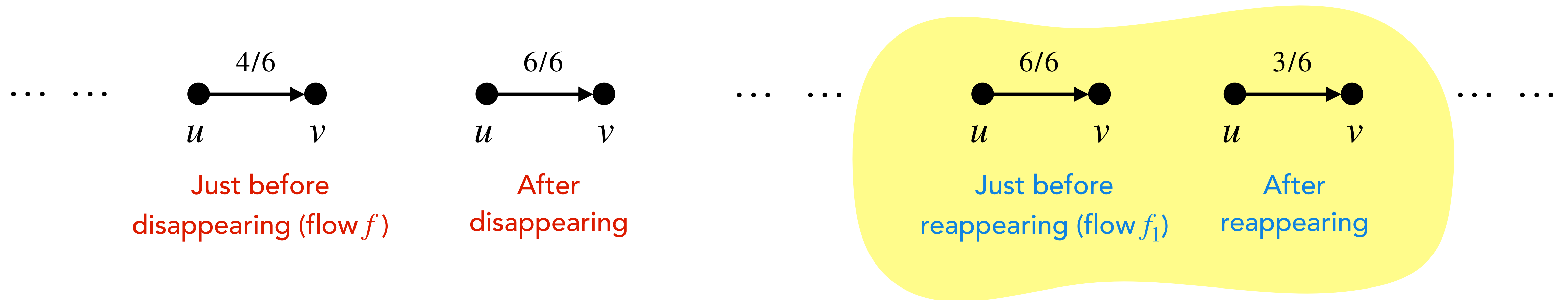
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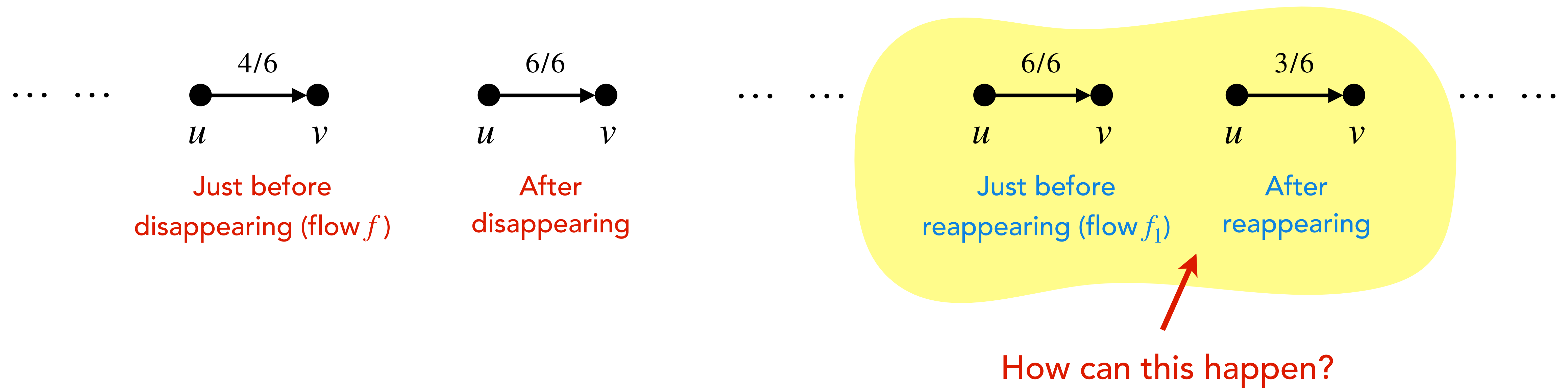
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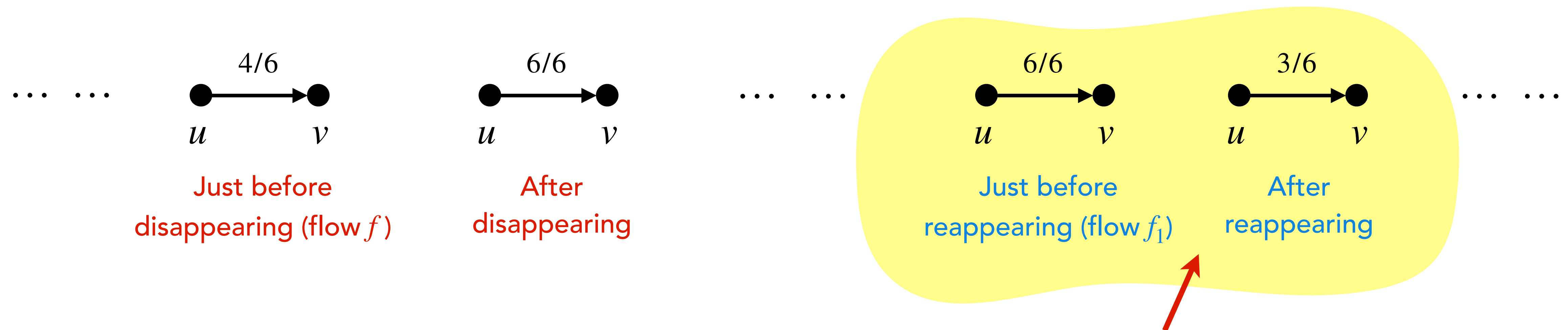
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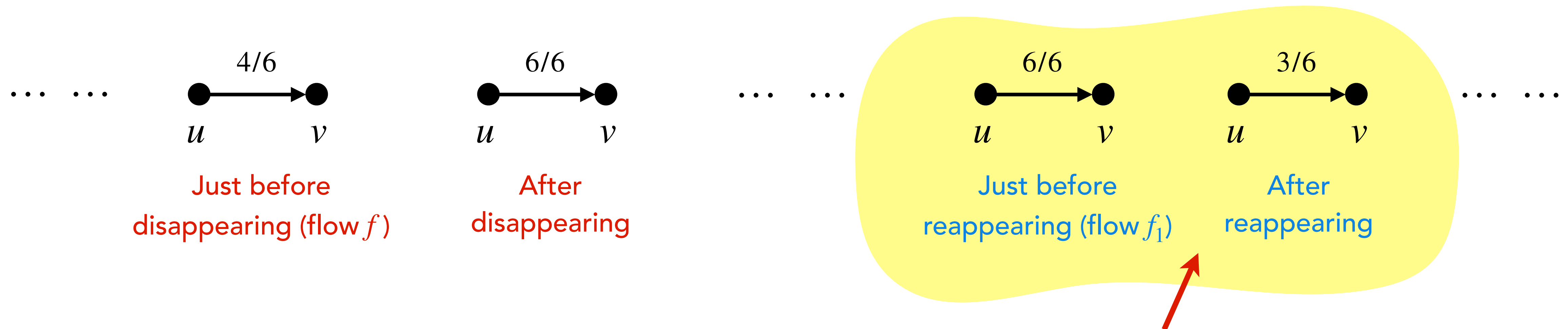
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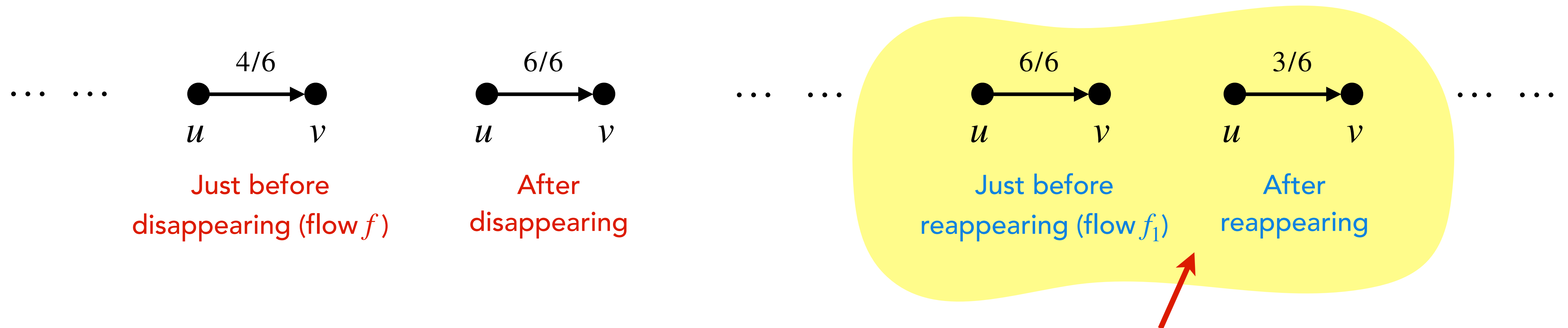
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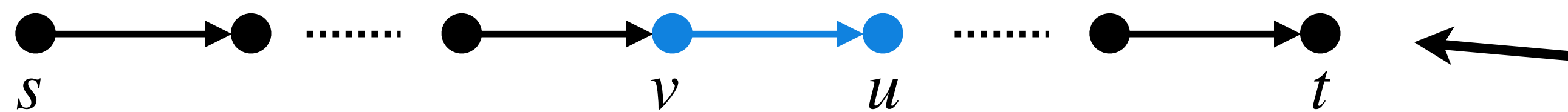
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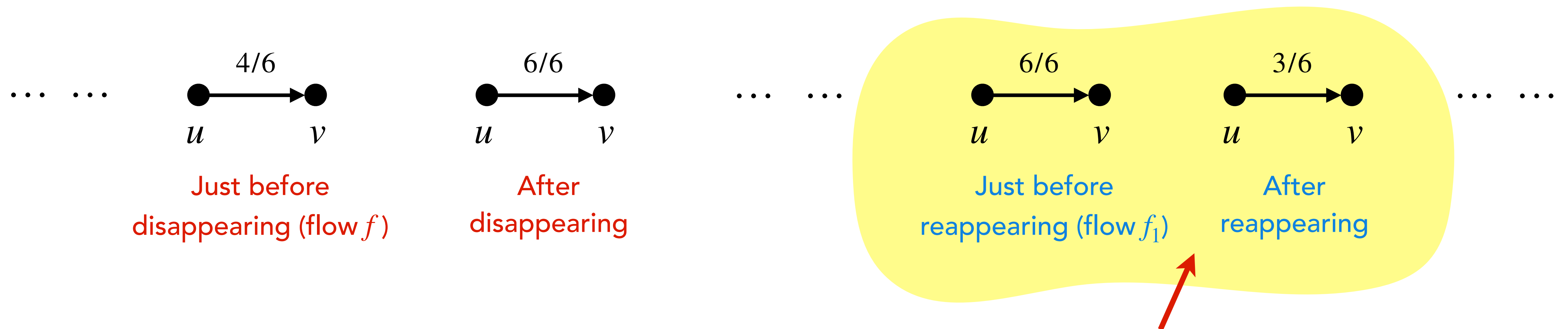
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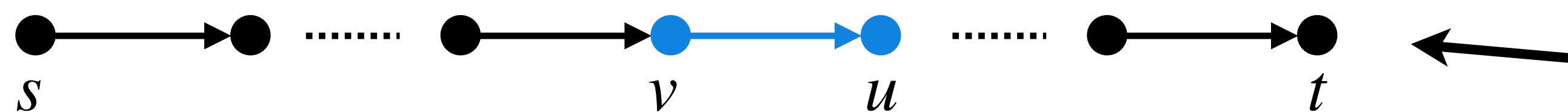
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- Hence, $\delta_{f_1}(s, u) = \delta_{f_1}(s, v) + 1$.



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Edmonds-Karp(G, s, t):

1. **for** each edge $(u, v) \in E(G)$
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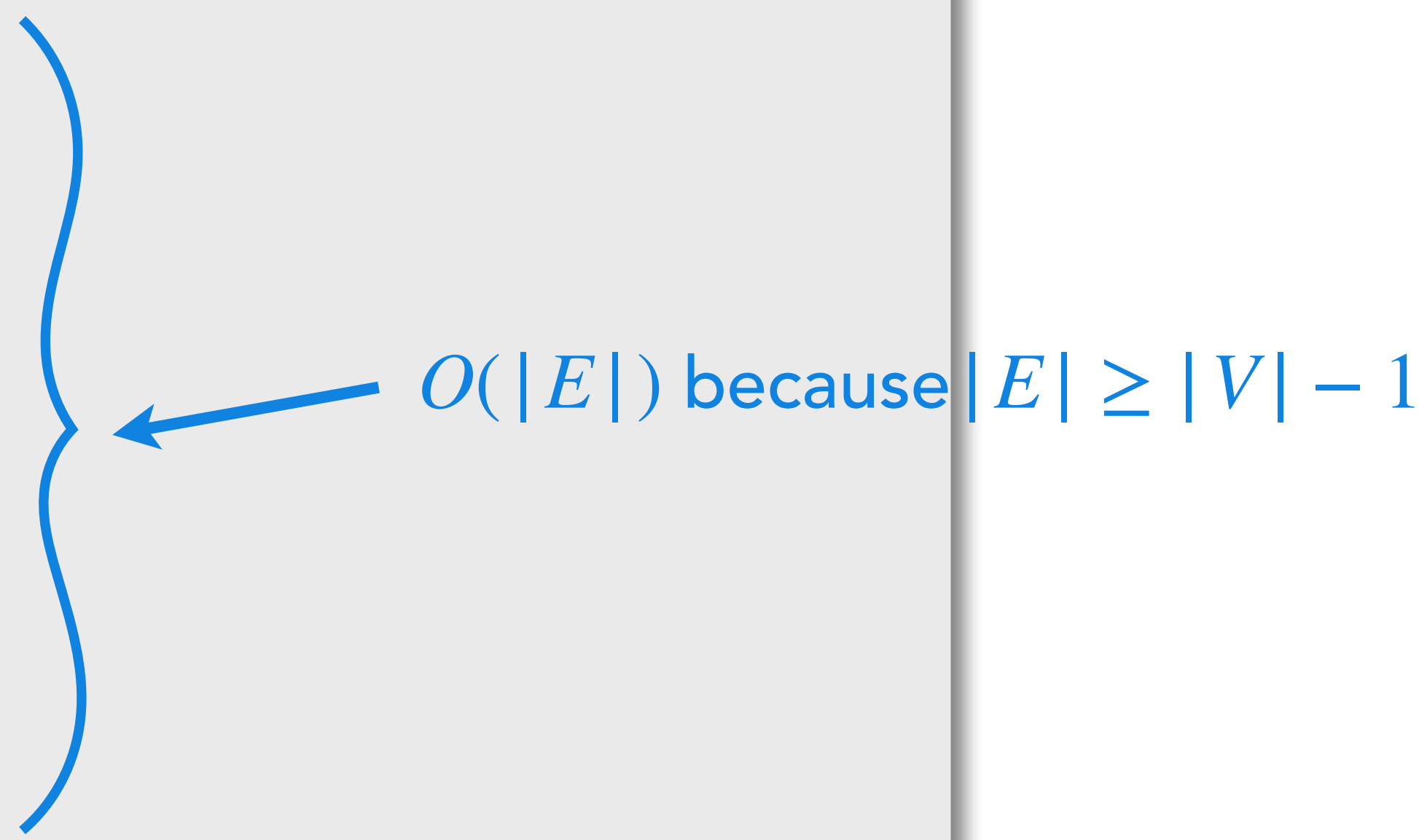
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Time Complexity: $O(|V| \cdot |E|^2)$